O2 Number theory — problems

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Problems are graded in three levels. Level 1 problems should take 2–5 minutes each, Level 2 problems 10–20 minutes, Level 3 problems can take a whole session. For every "find all" problem, prove that your list is complete. Write full sentences in proofs, as you would at the NWO second round.

Complete solutions are in solutions.md. One-line hints are at the bottom of this page.

Level 1 — Kangoeroe style (wizBRAIN)

1. *(number of divisors)* How many positive divisors does 720 have?

2. *(modular arithmetic)* What is the last digit of 7^{2028}?

3. *(modular arithmetic)* What is the remainder when 2^{100} is divided by 7?

4. *(lcm, remainders)* Find the smallest positive integer that leaves remainder 1 when divided by each of $2, 3, 4, 5$ and 6, and that is divisible by 7.

5. *(squares mod 10)* A perfect square (kwadraat) ends in one of the digits below. Which one can it be?

(A) 2 (B) 3 (C) 6 (D) 7 (E) 8

6. *(number bases)* When 2027 is written in base 2 (binary, tweetallig stelsel), how many of its digits are 1?

7. *(Legendre, trailing zeros)* How many zeros does $2027!$ end in?

8. *(gcd and lcm)* How many ordered pairs (a, b) of positive integers have $\gcd(a, b) = 12$ and $\operatorname{lcm}(a, b) = 360$?

9. *(prime factorisation)* What is the smallest positive integer n for which $2028n$ is a perfect cube (derde macht)?

10. *(divisibility rules)* a and b are digits such that the six-digit number $\overline{20a28b}$ is divisible by 36. How many pairs (a, b) are possible?

Level 2 — NWO first/second round style

11. *(Diophantine, SFFT)* Find all pairs of positive integers (x, y) with $xy + 2x + 3y = 30$.

12. *(Diophantine, factoring)* Find all pairs of positive integers (x, y) with $\dfrac{1}{x} + \dfrac{1}{y} = \dfrac{1}{6}$.

13. *(linear Diophantine)* A shop sells boxes of 7 and boxes of 11 pencils. In how many ways can you buy exactly 200 pencils? List them.

14. *(divisibility, Fermat)* Prove that $n^5 - n$ is divisible by 30 for every integer n.

15. *(gcd, Euclid)* For positive integers n, which values can $\gcd(n^2 + 3,\ n + 2)$ take? Give an example of each.

16. *(Fermat's little theorem)* Prove that $2^{70} + 3^{70}$ is divisible by 13.

17. *(primes, mod 3)* Find all primes p for which $2p^2 + 1$ is also prime.

18. *(number bases)* Find all bases $b \geq 2$ in which the number 2028 has exactly three digits and its last digit is 0.

19. *(digit sums)* Find all positive integers n such that $n + S(n) = 2027$, where $S(n)$ is the digit sum of n.

20. *(Diophantine, factoring)* Find all pairs of positive integers (x, y) with $x^2 - y^2 = 2028$.

21. *(number of divisors)* What is the smallest positive integer that has exactly 12 positive divisors? Prove that no smaller number works.

22. *(squares mod 5)* Find all positive integers n for which $1! + 2! + 3! + \cdots + n!$ is a perfect square.

Level 3 — NWO final / BxMO (Benelux Mathematical Olympiad) style

23. *(infinite descent)* Prove that the only integers x, y, z with $x^2 + y^2 = 3z^2$ are $x = y = z = 0$.

24. *(Diophantine, powers)* Find all pairs of positive integers (x, y) with $3^x - 2^y = 1$.

25. *(Fermat, order)* Find all positive integers n such that n divides $2^n - 1$.

26. *(powers of 2, divisibility)* Prove that for every integer $n \geq 2$ the number

$$H_n = 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n}$$

is not an integer.

27. *(Euclid's algorithm)* Prove that for all positive integers a and b,

$$\gcd(2^a - 1,\ 2^b - 1) = 2^{\gcd(a,b)} - 1.$$

28. *(Legendre, trailing zeros)* (a) Prove that there is no positive integer n for which $n!$ ends in exactly 2027 zeros. (b) Find all n for which $n!$ ends in exactly 2026 zeros.

29. *(Bézout, linear Diophantine)* Let a and b be coprime positive integers. Call a non-negative integer N *reachable* if $N = ax + by$ for some non-negative integers x, y.

(a) Prove that $ab - a - b$ is not reachable.
(b) Prove that every integer $N > ab - a - b$ is reachable.
(c) With stamps of 5 cent and 8 cent, what is the largest amount (in cents) you cannot pay exactly?

Hints

1. Factorise 720 into primes first.
2. Look at the last digits of $7, 7^2, 7^3, 7^4$.
3. $2^3 = 8 \equiv 1 \pmod 7$.
4. The number is 1 more than a multiple of $\operatorname{lcm}(2, 3, 4, 5, 6)$.
5. List $0^2, 1^2, \ldots, 9^2$ modulo 10.
6. $2047 = 2^{11} - 1$ is eleven 1s in binary, and $2027 = 2047 - 20$. Or divide by 2 repeatedly.
7. Legendre with $p = 5$.

8. Write $a = 12x$, $b = 12y$ with x, y coprime.
9. $2028 = 2^2 \cdot 3 \cdot 13^2$; every exponent must become a multiple of 3.
10. $36 = 4 \cdot 9$: last two digits and digit sum.
11. Add 6 to both sides and factor.
12. Clear denominators, then use SFFT.
13. Work modulo 7.
14. Check divisibility by 2, 3 and 5 separately; Fermat for 5.
15. Divide $n^2 + 3$ by $n + 2$ with remainder.
16. $3^3 \equiv 1$ and $2^{12} \equiv 1 \pmod{13}$.
17. What is $2p^2 + 1$ modulo 3 when $p \neq 3$?
18. Three digits means $b^2 \leq 2028 < b^3$; last digit 0 means b divides 2028.
19. How large can $S(n)$ be? Then look modulo 9.
20. Both factors $x - y$ and $x + y$ must be even.
21. $12 = 12 = 6 \cdot 2 = 4 \cdot 3 = 3 \cdot 2 \cdot 2$.
22. For $k \geq 5$, $k!$ ends in 0.
23. Squares modulo 3, then divide everything by 3.
24. Treat $y = 1$ separately; for $y \geq 2$ work modulo 4, then factor a difference of squares.
25. Let p be the smallest prime divisor of n and use the order of 2 modulo p.
26. Look at the largest power of 2 that is at most n.
27. Mimic one step of Euclid: $2^a - 1 = 2^{a-b}(2^b - 1) + (2^{a-b} - 1)$.
28. Find where $Z(n) = v_5(n!)$ jumps by more than 1.
29. (a) Suppose it is reachable and look modulo a and modulo b. (b) Look at $N, N - b, N - 2b, \ldots, N - (a - 1)b$ modulo a.